

Spreading 34 Points in the Plane

Limin Ge*

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Abstract

We place 34 points in the plane so that the squared ratio of the largest to the smallest distance between them is at most 30.90470+. Erich Friedman’s table of best known configurations lists n up to 31. Deleting a point from our 35-point configuration gives 32.07769; this one is smaller. The ratio is computed exactly from the published coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 34 points in the plane with*

$$\left(\frac{\max d}{\min d}\right)^2 = 30.90470\dots$$

The exact value, a rational number, is given in Appendix A.

2 Context

Erich Friedman’s page *Minimizing the Ratio of Maximum to Minimum Distance* lists the best known configurations for $n \leq 31$. The value for $n = 34$ is not on the page. Deleting a point from our 35-point configuration gives 32.07769; this one is smaller.

3 Verification

The coordinates are decimals with 12 digits after the point, so they are exact rationals; all squared distances, and their largest-to-smallest ratio, are computed with integers. Run

```
python friedman-maxmin2d/verify.py friedman-maxmin2d/coords/n34.txt
```

in <https://github.com/Ge-limin/math>.

4 How it was found

Random starts of a constrained local optimizer (SLSQP), then basin hopping seeded from the neighbouring sizes (`friedman-maxmin2d/`).

A Exact value

```
numerator:
15452354219017296516493805
denominator:
499999999999175627295293
```

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

B Coordinates

34 points, x and y per line. File: friedman-maxmin2d/coords/n34.txt.

-6.736633642682	-7.074012235443	-7.206502846312	-1.638987487038
-5.313555263900	-6.068090331541	-8.873165476042	-5.511461881327
-9.879053443371	-5.600695156999	-5.902761276102	-3.319807175734
-7.876441032073	-5.430589248172	-7.733208518099	-7.156707565942
-9.024057795447	-3.792663327237	-6.308309175480	-6.170387229815
-8.261203678961	-1.727948763830	-8.016838833840	-3.704059171728
-10.020782239417	-3.873535960391	-9.194301017938	-2.087572672926
-8.303112212795	-6.334996072307	-9.452382262649	-4.696288332865
-6.450210695175	-4.444898032208	-4.731136994815	-5.255201064890
-7.662958281470	-2.529261714169	-5.771026532627	-5.178865928880
-6.169020661707	-1.822101227331	-9.299836656764	-6.415868705461
-8.444841008840	-4.607836875699	-7.306537337377	-6.252300741808
-4.477054906749	-4.288018407620	-8.596055620447	-2.888885623265
-6.675095499552	-2.684590797413	-7.448116564871	-4.526964242544
-5.474983769822	-4.223691241929	-5.396686438257	-2.457317605653
-7.020114389871	-3.623186538573	-6.878212870175	-5.348675736180
-9.769419036055	-2.905643125416	-4.811821115514	-3.314345029463